Best Practice to Connect to an External Database

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Log 02: Additional information included pertaining to the usage of the Therefore™ Replication Service.

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1. What is the best option?

The purpose of this guide is to provide some best practice guidance on integrating Therefore™ with external databases. This white paper is based on Microsoft® SQL Server®, but similar principles may also be helpful for Oracle® databases (for Business and Enterprise Editions). We will look at the advantages and disadvantages of two options and then make some recommendations.

Creating a View

This is the typical and recommended method for installations where the Therefore™ Server can access the external database.

Therefore™ Table Replication

This method is primarily used for Therefore™ Online or other scenarios where the Therefore™ Server cannot directly access the external database.

See Comparision of View and Table Replication

Recommendations

Where the Therefore™ Server can access the external database directly:

- A View is recommended where records on the source database will not be deleted and documents should be retrievable with current record data.
- A View with Synchronized Redundant or Editable Redundant is recommended where records from the source could be deleted and/or documents should be retrievable with the index data from time of saving or manually edited since saving.
- Table Replication (with replicate deletions off) is recommended where records could be deleted from the source AND documents need to be retrieved with current record data. (Note: new documents can still be saved with the deleted record data.)

Where the Therefore™ Server cannot directly access the external database (e.g. Therefore™ Online):

- Table Replication (with replicate deletions on) is recommended where records on the source database will not be deleted and documents should be retrievable with current record data.
- Table Replication (with replicate deletions on) and Synchronized Redundant or Editable Redundant is recommended where records from the source could be deleted and/or documents should be retrievable with the index data from time of saving or manually edited since saving..
- Table Replication (with replicate deletions off) is recommend where records could be deleted from the source AND documents need to be retrieved with current record data. (Note: new documents can still be saved with the deleted record data.)
- Table Replication (with replicate deletions off) with Synchronized Redundant or Editable Redundant is not recommended.

2. Connection options

2.1 Creating a View

See the database manufacturer's documentation for details on creating a view.

- See the Appendix for an example on doing this for Active Directory.

2.2 Therefore™ Table Replication

Connection Requirements

- Microsoft SQL Server 2008+ and Oracle 11g R2+ are supported.
- Operating system where replication service is installed must be at least Windows Server 2008 R2 or Windows 7.
- Column type must be supported (see supported list below).

SQL Server (symbolic names):		Oracle:
Varchar Char Text NText NChar NVarchar Int1 Int2 Int4	Int8 DateTime DateTime4 Float4 Float8 Decimal Numeric Money Money4	Varchar2 Char NVarchar2 NChar Number Date Timestamp Clob NClob

Then, in order for the table replication to work, two connections are required.

- Firstly the Therefore™ Replication Service needs access to the database. The Therefore™ Replication Service must be installed and running on the source database or alternatively, a client or workstation that has 24/7 access to the source database.
- Secondly it needs access to Therefore™.

The table below summarizes the options for achieving this.

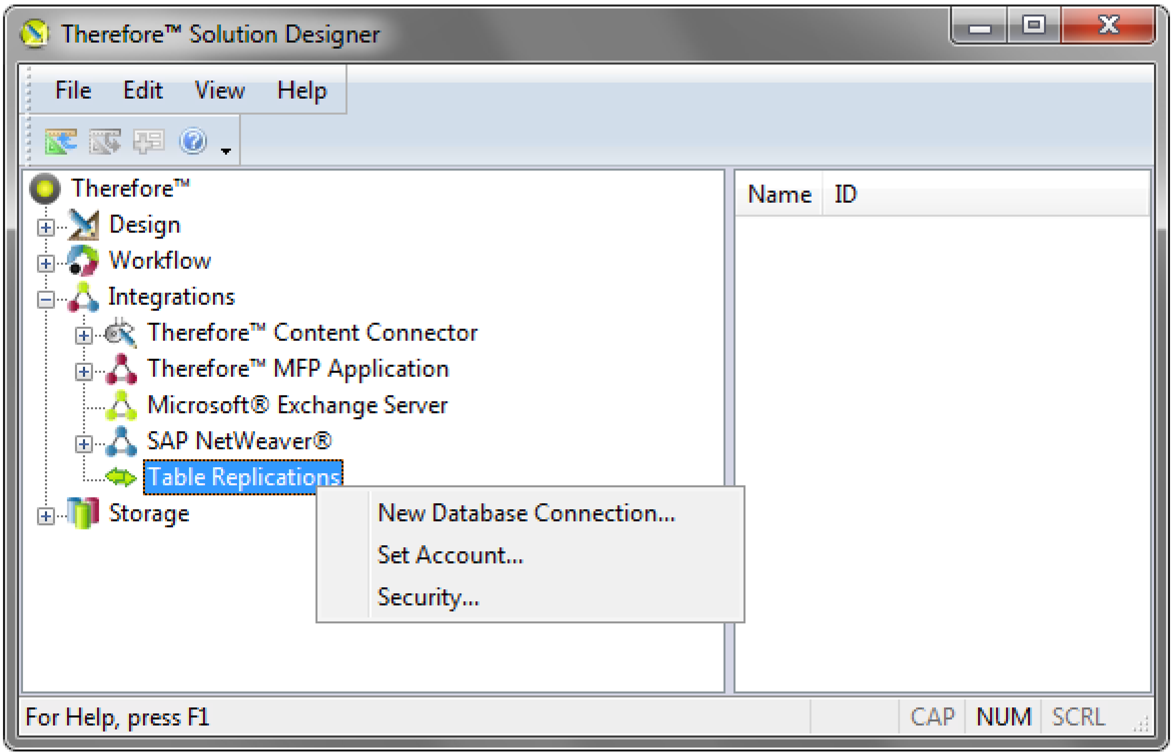
Account running the Therefore™ Replication Service	Does NOT have administrator rights on replicated table in Therefore™	Has administrator rights on replicated table in Therefore™
Has connect, create table, insert, update and delete rights on external database	DB Connection: Windows or DB authentication Set Account: Yes	DB Connection: Windows or DB authentication Set Account: No
Does NOT have rights on external database	DB Connection: DB authentication Set Account: Yes	DB Connection: DB authentication Set Account: No

In addition, the Windows account used to start the Therefore™ Solution Designer requires at least connect rights on the external database.

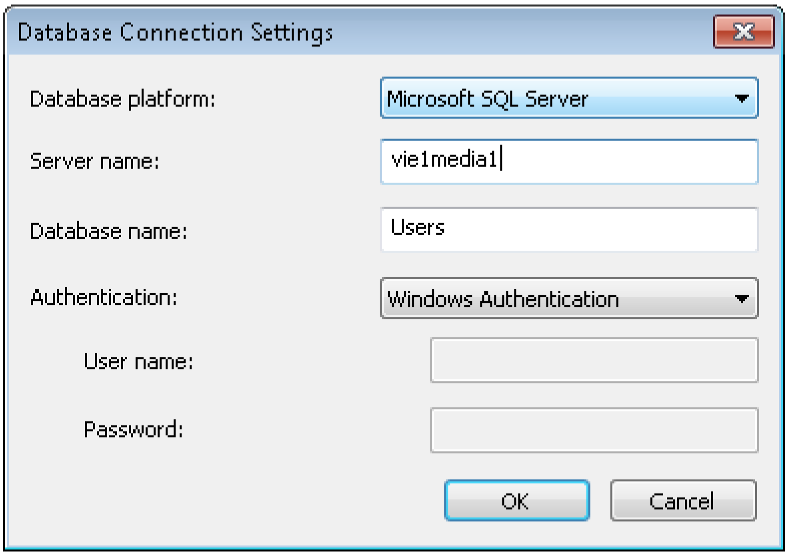
How do I create a table replication?

1. Please make sure you understand the Connection Requirements before proceeding.

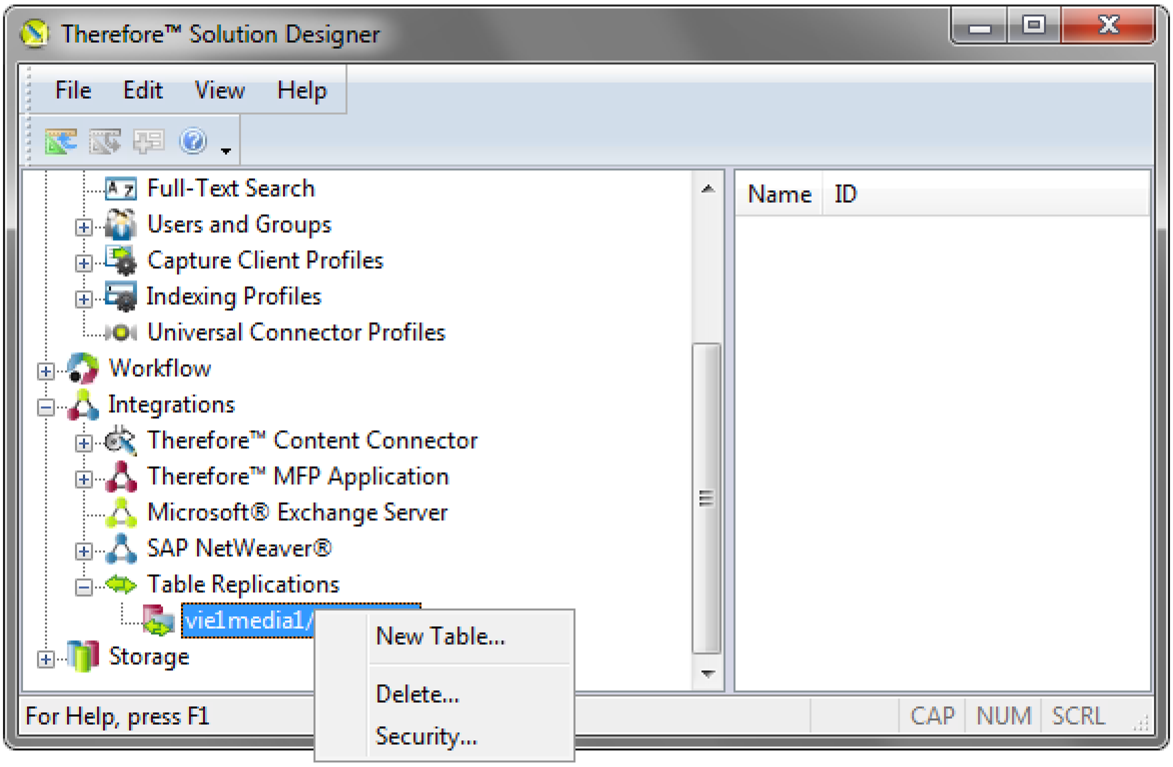
2. Open the Therefore™ Solution Designer and expand the Integrations object. Right-click on Table Replications and choose New Database Connection...



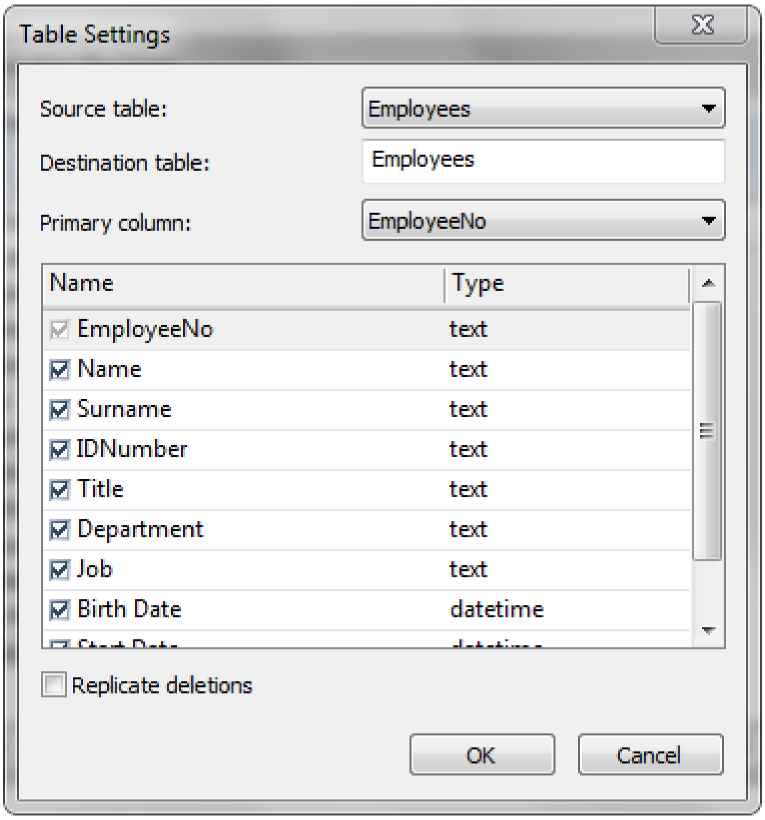
3. Select the Database platform, enter the Server name (or IP address) and the Database name. Then select the type of authentication. For database authentication, the user name and password must be specified. See the Connection Requirements for details on what authentication settings should be used and for details on how and when to use Set Account... Click OK to create the connection.



4. Right-click on the database connection that you have created and select New Table...



5. Select the Source table from the database. Under Destination table, enter an intuitive name for the replicated table and choose the Primary column for the table. This must be a unique column that does not contain NULL values. Then de-select any columns that you do not want to replicate and click OK to create the replicated table.



- You can set the interval for updates under the Advanced Server Settings (default is 4 hours).
- See the Appendix for an example on doing this for Active Directory.

3. Appendix

3.1 Example: Creating a View for Active Directory

1. The first step is to create a linked server:

   a) Set up ADSI mapping

```
EXEC sp_addlinkedserver 'ADSI', 'Active Directory Services 2.5',
'ADSDSOObject', 'adsdatasource'
GO
```

   b) Create a Microsoft SQL Server Authenticated Login

```
EXEC sp_addlinkedsrvlogin 'ADSI', false, 'MOYAWARE\Administrator',
'CN=Administrator, OU=Sales, DC=activeds, DC=Moyaware, DC=demo', 'password'
GO
```

Edit domain and user account login details.

- When defining the linked server, collation of the linked server MUST be set to the Microsoft SQL Server instance default collation.
- In case that Windows Authentication is used you may have to set the security of the linked server object ADSI to 'impersonate' for the user account starting the Therefore™ Server Service.
- The Therefore™ Server Service must not be running as the Local System account.

For more details refer to Microsoft online documentation:

http://msdn.microsoft.com/en-us/library/windows/desktop/aa772380%28v=vs.85%29.aspx
http://msdn.microsoft.com/en-us/library/windows/desktop/aa746475%28v=vs.85%29.aspx

2. The next step is creating a view to Active Directory.

```
CREATE VIEW dbo.VIEWADUsers
AS
SELECT     *
FROM          OPENQUERY(ADSI,
                       'SELECT name, adsPath, mail, sAMAccountName  FROM
''LDAP://DC=MOYAWARE, DC=DEMO'' WHERE objectCategory = ''Person'' AND
objectClass= ''user'' and mail = ''*'' ')
                        Rowset_1
GO
```

- Choose the Therefore database to execute this query statement.
- Edit domain details.

3.2 Example: Table Replication from Active Directory

1. The first step is to create a database where a view to Active Directory will be created. For a database name use a descriptive name like Users.

   For more details on how to create a database refer to Microsoft documentation: http://msdn.microsoft.com/en-us/library/ms186312%28v=sql.110%29.aspx

   - For this step you will need Microsoft SQL Server or Microsoft SQL Server Express installed locally.
   - An existing version of Microsoft SQL Server can also be used.

2. The next step is to create a linked server:

   a) Set up ADSI mapping

```
EXEC sp_addlinkedserver 'ADSI', 'Active Directory Services 2.5',
'ADSDSOObject', 'adsdatasource'
GO
```

   b) Create a Microsoft SQL Server Authenticated Login

```
EXEC sp_addlinkedsrvlogin 'ADSI', false, 'MOYAWARE\Administrator',
'CN=Administrator, OU=Sales, DC=activeds, DC=Moyaware, DC=demo', 'password'
GO
```

   Edit domain and user account login details.

   - When defining the linked server, collation of the linked server MUST be set to the Microsoft SQL Server instance default collation.
   - In case that Windows Authentication is used you may have to set the security of the linked server object ADSI to 'impersonate' for the user account that is used for the authentication.

   For more details refer to Microsoft documentation:

   http://msdn.microsoft.com/en-us/library/windows/desktop/aa772380%28v=vs.85%29.aspx
   http://msdn.microsoft.com/en-us/library/windows/desktop/aa746475%28v=vs.85%29.aspx

3. The next step is creating a view to Active Directory. Depending on what users need to be fetched from the Active Directory different views can be created.

   For fetching all Active Directory users use the following script:

```
CREATE VIEW dbo.VIEWADUsers
AS
SELECT     *
FROM         OPENQUERY(ADSI,
                      'SELECT name, adsPath, mail, sAMAccountName  FROM
''LDAP://DC=MOYAWARE, DC=DEMO'' WHERE objectCategory = ''Person'' AND
objectClass= ''user'' and mail = ''*'' ')
                       Rowset_1
GO
```

For fetching only Active Directory users that are members of certain groups use the following script:

```
CREATE VIEW dbo.VIEWADUsers
AS

SELECT     'Customer Service' AS Department, *
FROM         OPENQUERY(ADSI,
                      'SELECT displayName, adsPath, mail, sAMAccountName  FROM
''LDAP://DC=Moyaware,DC=com''
   WHERE objectCategory = ''Person''
   AND objectClass= ''user''
   AND mail = ''*''
   AND memberOf = ''CN=CustomerService,CN=Users,DC=Moyaware,DC=com'' ')
                       Rowset_1

UNION
SELECT     'HR' AS Department, *
FROM         OPENQUERY(ADSI,
                      'SELECT displayName, adsPath, mail, sAMAccountName  FROM
''LDAP://DC=Moyaware,DC=com''
   WHERE objectCategory = ''Person''
   AND objectClass= ''user''
   AND mail = ''*''
   AND memberOf = ''CN=HR,CN=Users,DC=Moyaware,DC=com'' ')
                       Rowset_1

UNION
SELECT     'Finance' AS Department, *
FROM         OPENQUERY(ADSI,
                      'SELECT displayName, adsPath, mail, sAMAccountName  FROM
''LDAP://DC=Moyaware,DC=com''
   WHERE objectCategory = ''Person''
   AND objectClass= ''user''
   AND mail = ''*''
   AND memberOf = ''CN=Finance,CN=Users,DC=Moyaware,DC=com'' ')
                       Rowset_1

GO
```

- Choose the Users database to execute this query statement.
- Edit domain and group details.
- It is possible to fetch as many groups as needed

4. For the table replication a non-null column with unique values need to exist. This column will be used as primary column during the table replication. Create a view that contains such a column by using the following script:

```
CREATE VIEW dbo.ADUsers
AS

SELECT ISNULL(ROW_NUMBER() OVER(ORDER BY mail), 0) AS RowNo, *
FROM dbo.VIEWADUsers
```

5. Create table replication by following the steps described under Therefore™ Table Replication. For primary column select RowNo.

3.3 Dependency Mode

Primary and dependent fields can be added to Therefore™ categories. Dependent fields can be set to one of three modes:

Dependency Mode:

Referenced
This is the default setting. The field value is fetched from the referenced table.

Editable Redundant
If this is selected, a snapshot of this column is copied to the Therefore™ database and the field is not updated if changes to the referenced table are made. If this field is changed, the other dependent fields and the primary field will not be updated.

Synchronized Redundant
If this is selected, a snapshot of this column is copied to the Therefore™ database and the field is not updated if changes to the referenced table are made. However, if this field is changed, the other dependent fields and the primary field will be updated from the referenced table.

Dependency mode may also be important in the scenario where a database field is needed for creating a condition in a Workflow condition. Only index fields that are directly saved in the Therefore™ database are listed and can be used in conditions. This means that primary fields are listed, but dependent fields set to Referenced are not. If the condition cannot be built with the primary field, the dependency mode could be set to Synchronized Redundant or Editable Redundant.

Note: Dependency mode can be set in the Dependency tab under Field Properties in the Therefore™ Solution Designer. This option is available only for dependent fields.

3.4 Comparision of View and Table Replication

The following table compares these two options including the use of the dependency mode in Therefore™.

Consideration	View	Table Rep.	View + Redundant	Table Rep. + Redundant
Are new documents saved with current record data from source database?	Yes	Yes[1]	Yes	Yes[1]
Are documents retrievable with index data from time of saving, after a record is changed in the source?	No	No	Yes	Yes
Is a document's index data updated after a record is changed in the source?	Yes	Yes	No	No
Are documents retrievable after record deletion?	No	Yes[2]	Yes	Yes
Can dependent index fields be used for workflow transitions?	No	No	Yes	Yes

Consideration	View	Table Rep.	View + Redundant	Table Rep. + Redundant
Can it be used with Therefore™ Online?	No	Yes	No	Yes
What is saved in the Therefore™ database?	View definition	All selected columns	View definition + saved index data	All selected columns + saved index data

1 Minimum update interval is 1 minute. Changes are always replicated, but when replicate deletions is turned off (default), records deleted in the source table will not be deleted from the replicated table. Hence new documents can still be saved with this deleted data.

2 Documents are only retrievable when replicate deletions is turned off (default).